1.1 Open and Closed Sets

A set $E \subseteq R$ is said to be an **open** set if, for any $x \in E$ there exists an open interval O containing x such that $O \subseteq E$.

Proposition 1.1.1: R and $\emptyset$ are open sets. *Proof*: Obvious. **QED**

Clearly, any open interval (a, b) is open.

Proposition 1.1.2: The union of any collection of open sets is open. The intersection of any finite collection of open sets is open.

Proof: Let $O = \bigcup O_i$ be the union of collection of open sets $\{O_i\}_{i \in I}$, where I is some index set. We would like to show O is open. Consider any $x \in O$. There exists some O_i such that $x \in O_i$. Since O_i is open, there exists some neighbor O' of x such that $O' \subseteq O_i$. As a result, $O' \subseteq O$. Therefore, O is open.

Let $F = \bigcap_{j=1}^{n} O_j$ be the intersection of a finite collection of open sets. We show F is open. To this end, consider any $x \in F$. Then $x \in O_j$ for $j = 1, 2, \ldots, n$. Since these O_j are all open, there are $r_1, \ldots, r_n > 0$ such that $(x - r_j, x + r_j) \subseteq O_j, j = 1, 2, \ldots, n$. Take $r = \min\{r_1, \ldots, r_n\}$. So $(x - r, x + r) \subseteq O_j$ for any j. Thus, $(x - r, x + r) \subseteq F$ and F is open. **QED**

Proposition 1.1.3: A non-empty open set is a countable union of disjoint open intervals. *Proof:* Let O be a non-empty open set. Consider $x \in O$. Denote by E the set whose member is a neighbor of x and is contained in O. Since O is open, E is not empty. Note that a member of E can be donoted as (a_i, b_i). Let $a_x = \inf\{a_i\}$ and $b_x = \sup\{b_i\}$. Let $I_x = (a_x, b_x)$. Thus, for each x, we construct an associated I_x. It is not difficult to show $I_x \subseteq O$.We claim that for $x \neq y$, either $I_x = I_y$ or I_x, I_y are disjoint.

To see this, suppose I_x, I_y are not disjoint and $a_x \neq a_y$. We may assume $a_x < a_y$. In this case, $(a_x, b_y) \in E_y$, i.e. (a_x, b_y) contains y and is contained in O. As a result, it must be the case $a_y \leq a_x$, which leads to a contraction. So $a_x = a_y$. By a similar argument, we can show $b_x = b_y$.

Now let $I = \{I_x\}_{x \in O}$. Clearly, $O = \bigcup I_x$. For countability, note that I_x contains a rational number. So $I \preceq Q$ and thus I is countable. **QED**

A point x is said to be a **point of closure** of a set E if, for any $r > 0$, $(x - r, x + r) \cap E$ is not empty.

Given a set E, we denote by $\bar{E}$ the set of points of closure of E.

A set E is said to be a **closed set** if it contains all its closure, i.e. $E = \bar{E}$.

Proposition 1.1.4: For any set E, its closure $\bar{E}$ is a closed set. *Proof:* Let x be a point of closure of $\bar{E}$. We need to show $x \in \bar{E}$. To this end, notice that there exists $a \in (x - r, x + r) \cap \bar{E}$. So a is a point of closure of E . Then there exists $r_1 > 0$ such that $(a - r_1, a + r_1) \subset (x - r, x + r)$ and $(a - r_1, a + r_1) \cap E \neq \emptyset$. Therefore, $(x - r, x + r) \cap E \cap \emptyset$. Hence, x is a point of closure of E. **QED**

Proposition 1.1.5: A set is closed if and only if its complement is open. *Proof:* Suppose set E is closed. Let O be $R - E$. Consider $x \in O$. Then x is not a closure of E. Thus there exists $r > 0$ such that $(x - r, x + r) \cap E = \emptyset$. That is, $(x - r, x + r)$ is contained in O. Hence, O is open.

Conversely, supppose O is open. Any point of closure of E cannot be in $$O$. Therefore, E is closed. **QED**

Proposition 1.1.6: R and $\emptyset$ are both closed sets. The intersection of any collection of closed sets is closed. The union of any finite collection of closed sets is closed. *Proof:* Since R and $\emptyset$ are open, by Proposition 1.1.5 they are also closed. Let $F = \{F_i\}$ be a collection of closed sets. The complement of the intersection is $R - \bigcap F_i = \bigcup(R - F_i)$. Since $R - F_i$ is open, $\bigcup(R - F_i)$ is open. Therefore, $\bigcap F$ is closed.

If F is finite, $\bigcap(R - F_i)$ is open and thus $\bigcup F_i$ is closed. **QED**

Now we present an important theorem in real analysis.

Theorem 1.1.7 (Heine-Borel Theorem): Any open cover of a non-empty closed and bounded set has a finite sub-cover. *Proof:* First consider a clsoed and bounded interval $[a, b]$. Let $O = \{O_i\}$ be a open cover. Now define a set E. If $x \in E$, then $a \leq x \leq b$ and $[a, x]$ has a finite sub cover. It is easy to show that E is not empty.

Since E has a upper bound, by Completeness Axiom E has a supremum c. If $c = b$, we are done. For $c \in [a, b]$, there must be a O' such that $c \in O'$. Then there is a r such that $(c - r, c + r) \subseteq O'$. Take any $x \in (c - r, c)$. Then $[a, x]$ has a finite sub-cover $O_{i_1}, \ldots, O_{i_k}$. Thus, $O_{i_1}, \ldots, O_{i_k}, O'$ is a finite sub-cover of $[a, c]$, i.e. also a finite sub-cover of $[a, b]$.

To show $c = b$, we essentially repeat the above argument. Assume $c < b$. By a similar argument, $[a, c]$ has a finite sub-cover. In addition, this finite sub-cover is also a cover of $[a, c']$, where $c < c' < b$. Therefore, $c' \in E$, which contradicts the fact that c is a supremum of E.

Next, we consider an arbitrary non-empty closed and bounded set F. Since F is bounded, there exists $M > 0$ such that $F \subseteq [-M, M]$. Let O be an open cover of F. Then $O \cup \{R - F\}$ is an open cover of $[-M, M]$. Thus, there is a finite sub-cover $O_{i_1}, \ldots, O_{i_m}$. By possibly removing $R - F$ from this finite sub-cover, we obtain a finite sub-cover of F. **QED**

In the proof of Heine-Borel theorem, we used the Completeness Axiom. This means Completeness Axiom $\Rightarrow$ Heine-Borel Theorem. The converse is also true.

Proposition 1.1.8: Heine-Borel Theorem $\Leftrightarrow$ Completeness Axiom. *Proof:* The $\Leftarrow$ part is shown in the proof of Theorem 1.1.7. For the $\Rightarrow$ part, consider a set F which is upper bounded. We would like to show that F has a supremum. To this end, consider its closure $\bar{F}$. We construct an open cover of $\bar{F}$ as follows.

If $\bar{F}$ has a maximum, say b, then it is not difficult to show that b is the supremum of F. For every $x \in F$ satisifies $x \leq b$, and any open set $(b - r, b)$ has non-empty intersection with F since b is a point of closure of F.

Now if $\bar{F}$ does not has a maximum, then we can construct an open cover $\{(-\infty, x)\}_{x \in \bar{F}}$. Note this is indeed an open cover of $\bar{F}$. Because for any $y \in \bar{F}$, there exists a $x > y$ such that $y \in (-\infty, x)$. Now by Heine-Borel Theorem, there is a finite sub-cover $(-\infty, x_1), \ldots, (-\infty, x_k)$ that covers $\bar{F}$. It also covers F. Let $x' = \max\{x_1, \ldots, x_k\}$. Then x' is the supremum of F. Because, again, every $x \in F$ satifies $x < x'$ and $x \in \bar{F}$ is a point of closure of F. **QED**

Next, we present a consequence of Heine-Borel Theorem. In fact, this resut is also equivalent to Heine-Borel Theorem.

Theorem 1.1.9 (Nested Set Theorem): $\{F_n\}_{n=1}^{\infty}$ is a descending countable collection of non-empty closed sets. F_1 is bounded. Then $\bigcap F_n$ is not empty. *Proof:* Suppose $\bigcap F_n$ is empty. Let $O_n = R - F_n$. Then $O_1, \ldots, O_n, \ldots$ is an ascending sequence of open sets. Note $\bigcup O_n = R - \bigcap F_n = R$. Since F_1 is closed and bounded, and $O_1, \ldots, O_n, \ldots$ is clearly an open cover of F_1, there is a finite sub-cover $O_{n_1}, \ldots, O_{n_k}$ of F_1. It is also a cover of F_{n_k}. Since $O_{n_1}, \ldots, O_{n_k}$ is ascending, we have O_{n_k} covers F_{n_k}, which is a contradiction. **QED**

Proposition 1.1.10: Nested Set Theorem $\Leftrightarrow$ Completeness Axiom. *Proof:* The $\Leftarrow$ part is taken care of in the proof of Heine-Borel Theorem and Nested Set Theorem. For the $\Rightarrow$ part, let E be a set that has a upper bound b. If E has a maximal element, we are done. Otherwise, take any $x \in E$. Then there are infinitely many $y \in E$ such that $x < y$. We construct a descending coutable sequence of closed sets $F_n = [a_n, b_n]$ as follows.

Let F_1 be $[x, b]$. Take $c = \frac{x+b}{2}$. If c is a upper bound of E, then let $a_2 = x$ and $b_2 = c$. Otherwise, let $a_2 = c$ and $b_2 = b$. Continuing this process, we obtain a sequence $F_n = [a_n, b_n]$. Note this sequence posses two properties:

- (a) Every $[a_n, b_n]$ contains at least one upper bound of E.
- (b) Every $[a_n, b_n]$ contains at least one element of E.
- (c) For any $y > x$, there is a $[a_n, b_n]$ such that $y < a_n$.

Property (a) and (b) can be proved by induction. For property (c), suppose $y \in [a_n, b_n]$ for all n. Since there is a $z > y$ and $b_n - a_n$ is decreasing, there exists N such that $b_N - a_N < z - y$. Note b_N is a upper bound of E. So $a_N \leq y < z < b_N$, which is a contradiction.

Now, by Nested Set Theorem, there exists $d \in \bigcap_{n=1}^{\infty} [a_n, b_n]$. d must be a upper bound of E. Otherwise, by property (c) there exists $[a_k, b_k]$ such that $d < a_k$, which contradicts the fact that $d \in [a_k, b_k]$. Finally, d must be a supremum of E. For otherwise, there is another upper bound $d' < d$. Since d is in every $[a_n, b_n]$, there is a $[a_N, b_N]$ such that $d' < a_N \leq d \leq b_N$. Thus, $[a_N, b_N]$ contains no element from E, which is a contradiction of property (b). **QED**